

Cybersecurity Incident Report:

Network Traffic Analysis

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Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP protocol reveals that udp port 53 is unreachable when attempting to access it.

This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message udp port 53 unreachable.

The port noted in the error message is used for: sending a query to the DNS server to find the corresponding IP address.

The most likely issue is ICMP flooding.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

Time incident occurred at 13:24:32.192571

Explain how the IT team became aware of the incident: We received complaints from clients unable to reach the website.

Explain the actions taken by the IT department to investigate the incident: We tried to verify by trying to access the website ourselves and found that it was not reachable. We loaded the network analyzer tool, tcpdump, and attempted to load the website again.

Note key findings of the IT department's investigation (i.e., details related to the port affected, DNS server, etc.): The analyzer revealed that when we sent UDP packets to the DNS server, we received ICMP packets containing a udp port 53 error message saying it is unreachable, we tried a number of times and received the same error message.

Note a likely cause of the incident: The most likely cause of this incident could be a threat actor using an ICMP flooding method. This means the actor sent a large volume of ICMP echo packets to overwhelm the DNS server.

